

CMP682 — Final Project

MissVARPath

Missense Variant Pathogenicity Prediction

Doruk Topcu · Alihan Sağöz

Hacettepe University · 2026

Three questions

Every missense variant goes through a battery of in-silico predictors — PolyPhen-2, SIFT, REVEL, CADD, AlphaMissense, DITTO. Each disagrees with the others; most were trained on overlapping ClinVar data — the same database we use to label our targets.

- 1.** Can a meta-classifier do better than any single predictor by combining them?
- 2.** How much of any improvement is data-circularity — predictors trained on ClinVar, evaluated on ClinVar?
- 3.** Can we predict pathogenicity from raw data alone — without using any pre-existing predictor's score?

Data

21,872

ClinVar / OpenCRAVAT missense variants

5,468 × 4

perfectly balanced 4-class (Benign / Likely-benign / Likely-pathogenic / Pathogenic)

777 → 208

raw schema → numerical features after preprocessing

5,428

strict-VUS rows in the curated pro-set (clinvar__sig = “Uncertain significance”)

Two task definitions

4-class

the original ClinVar label space

2-class

Likely-* collapsed into definitive labels → 10,936 / 10,936

3-class (with VUS)

Benign-side / Pathogenic-side / VUS

5-class (with VUS)

all four ClinVar labels + VUS — essentially balanced

Methods

Model suite — 11 estimators

Classical / linear

- KNN
- NearestCentroid
- CosineSimilarity (custom)
- DecisionTree
- LDA
- QDA
- LinearSVC
- RidgeClassifier
- SGDClassifier

Ensembles

- AdaBoost
- HistGradientBoosting

Evaluation regimes — 5 cuts

Canonical	stratified 80/20 holdout + 5-fold CV
Augmented	+ 196 k-mer + 10 BLAST locus-neighbour features
Gene-stratified	no gene appears in both train and test folds
No-VEP / VUS	drop every learned-predictor + conservation score
Raw-only	keep only population AF, position, sequence composition

RANDOM_STATE = 42 throughout. Per-model grid-search tuning by 5-fold CV macro-F1 on the training portion; AdaBoost grid stopped at combo 23/27 (compute budget).

Headline numbers

4-class macro-F1

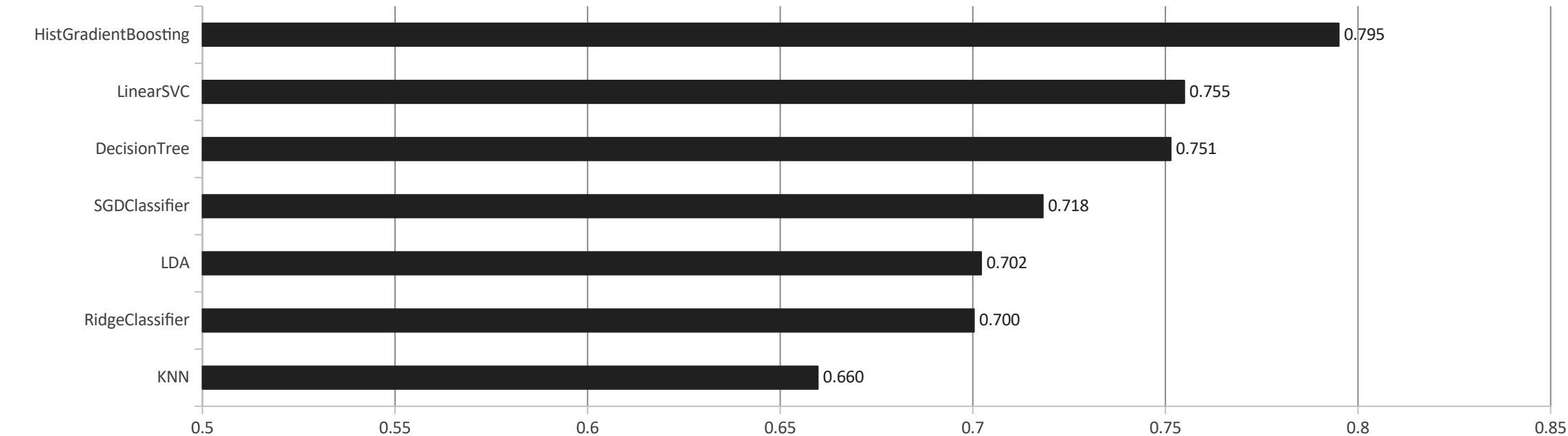
0.795

HistGradientBoosting · 21,872 variants · 208 features

2-class macro-F1

0.987

HistGradientBoosting · 10,936 / 10,936 collapse

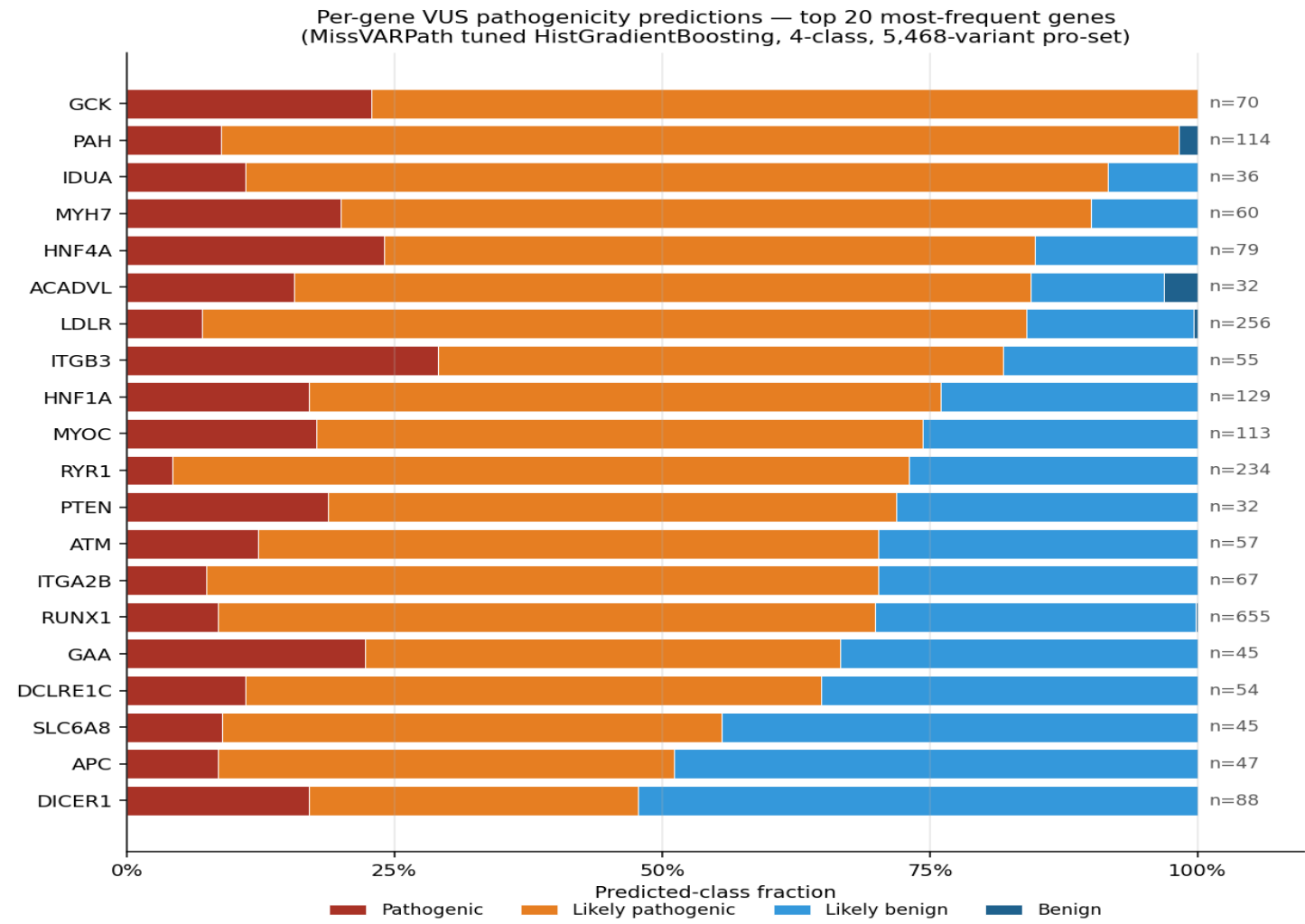


Stress tests — bounding the data-circularity

Regime	4-class F1	2-class F1
Canonical tuned (headline)	0.795	0.987
Augmented (k-mer + BLAST)	0.795	0.990
Gene-stratified CV	0.766	0.987
No-VEP, k-fold	0.773	0.976
No-VEP, gene-stratified	0.761	—
Raw-only (no predictor, no conservation)	0.721	0.934
DITTO removed	0.791	—

Graceful degradation across regimes: HistGB loses 0.027 to gene-stratified CV, 0.022 to no-VEP, 0.074 to raw-only. The 2-class task is essentially insensitive to all stresses — that's the operationally-deployable headline.

Real VUS deployment



On the 5,468 strict-VUS pro-set

62% / 38%

pathogenic-side / benign-side split on 2-class head

80%

of predictions have max probability > 0.9

87%

of 4-class predictions hedge into Likely-* — only 13% definitive

95.4%

cross-task agreement (4-class collapsed vs. 2-class head)

GCK 100% · PAH 99% · MYH7 90% · LDLR 84% — clinically sensible per-gene priors that the model could only have learned indirectly through gene-correlated features.

VUS as a class

With VUS added as a labelled class

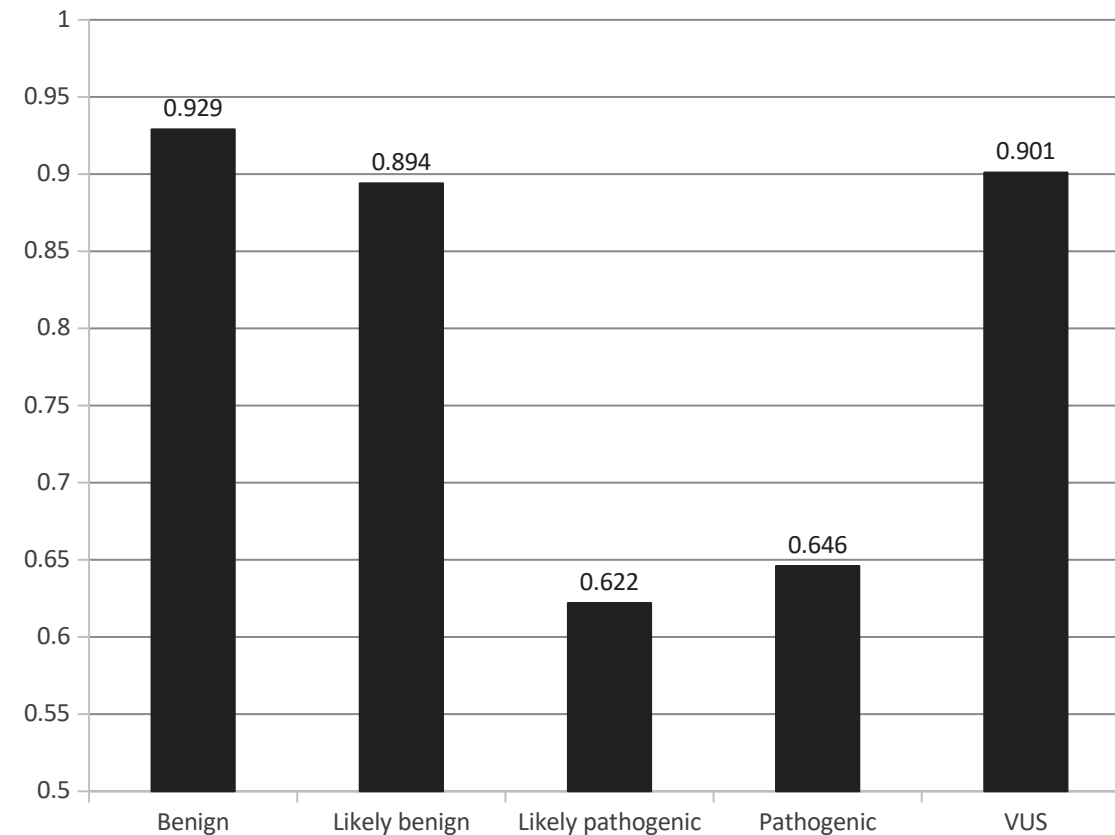
2-class no-VUS **0.987**

3-class with VUS **0.946**

4-class no-VUS **0.795**

5-class with VUS **0.798**

Adding VUS as a 5th class does NOT hurt overall macro-F1 — it slightly improves it. MCC improves by +0.022.



VUS reaches F1 = 0.901 — between Benign and Likely-benign in difficulty. One of the cleanest classes to recognise.

Useful vs. waste — feature ablation

Group	4-class	2-class	5-class	3-class	n feat
population_af	+0.0471	+0.0025	+0.0405	+0.0054	47
functional	-0.0012	-0.0005	+0.0266	+0.0469	4
other_vep	+0.0133	-0.0002	+0.0137	+0.0073	56
cadd	-0.0054	-0.0005	-0.0001	-0.0015	4
ditto	+0.0039	+0.0046	+0.0001	+0.0033	1
metarnn	+0.0014	+0.0005	+0.0045	+0.0005	2
bayesdel	-0.0045	-0.0002	-0.0004	-0.0003	4
position	-0.0044	+0.0002	+0.0018	+0.0029	2
revel	-0.0018	+0.0005	-0.0044	-0.0015	2
chasmplus	+0.0026	+0.0011	+0.0028	-0.0002	68
alphamissense	-0.0015	+0.0005	-0.0028	-0.0017	1
conservation	+0.0018	-0.0007	-0.0001	+0.0000	17

But — Phase B per-feature permutation

ditto_score

0.133 individual macro-F1 weight (4-class)

metarnn_score

0.048 individual macro-F1 weight (4-class)

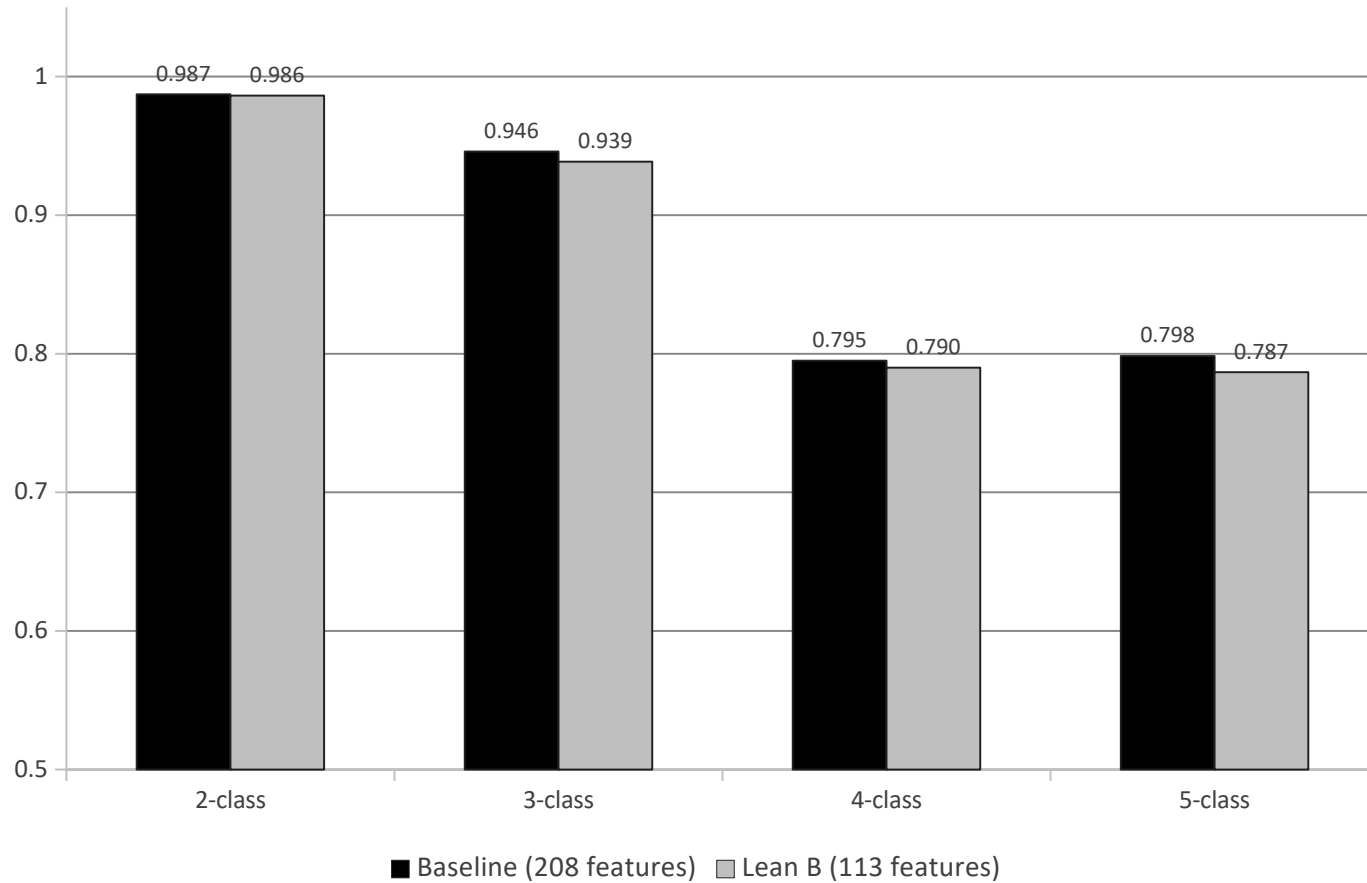
The resolution

Inter-predictor redundancy. Remove DITTO and MetaRNN compensates; remove MetaRNN and DITTO compensates. Group LOO measures replaceability, not informativeness.

Greyed rows: HistGB $|\Delta| < 0.005$ across every task. 8 of 12 groups (97 of 208 features) are waste at the group level.

Lean B & future directions

Lean B — drop 95 of 208 features (45%), keep ditto_score + metarnn_score



At most 0.012 macro-F1 cost on any task. Several models actually improve.

Future directions

- **Cross-database validation**

replicate on HGMD or a held-out ClinVar snapshot

- **True homology BLAST**

blastp on UniRef50 with the AA substitution applied

- **Probability calibration**

Platt / isotonic for usable risk scores

- **Gene-stratified VUS-as-class**

test whether VUS-detection survives without same-gene leakage

- **Variant-level error analysis**

manual review of LP $\leftrightarrow$ P confusion region